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A Project Report on

"HELMET DETECTION USING YOLO V8"

Submitted in partial fulfilment for the award of the degree of

BACHELOR OF TECHNOLOGY
IN
COMPUTER SCIENCE AND ENGINEERING
(ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING)

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ABSTRACT

Road safety has become a major concern worldwide due to the rapid increase in the number of vehicles, especially two-wheelers. Among various safety measures, wearing a helmet is one of the most effective ways to reduce the risk of severe head injuries and fatalities during accidents. Despite strict traffic regulations, many riders fail to comply with helmet rules, mainly due to inadequate monitoring and enforcement systems. Traditional methods of traffic monitoring rely heavily on manual observation by traffic personnel or basic CCTV surveillance, which are inefficient, time-consuming, and prone to human error.

This project proposes an automated helmet detection system using *YOLOv8 (You Only Look Once version 8)*, a state-of-the-art deep learning algorithm for real-time object detection. The system is designed to detect two-wheelers and identify whether the rider is wearing a helmet or not using video streams or CCTV footage. A large dataset consisting of helmet and non-helmet images is used to train the model. The images are preprocessed and annotated in YOLO format to improve detection accuracy.

The trained model is capable of processing video frames in real time, detecting riders, and classifying them based on helmet usage. The system can also capture images of violators and store them for further action. Performance metrics such as accuracy, precision, recall, and mean Average Precision (mAP) are used to evaluate the model's effectiveness.

The proposed system offers a scalable and efficient solution for traffic rule enforcement and can be integrated into smart city infrastructure. It reduces the dependency on manual monitoring and enhances road safety by ensuring better compliance with helmet laws.

CHAPTER 1

INTRODUCTION

1.1 Background

With the rapid growth in population and urbanization, the number of vehicles on roads has increased significantly. Among these, two-wheelers are one of the most commonly used modes of transportation, especially in countries like India due to their affordability and convenience. However, this increase in two-wheeler usage has also led to a rise in road accidents and fatalities. One of the major causes of severe injuries in such accidents is the absence of proper safety measures, particularly the non-usage of helmets by riders.

Helmets play a crucial role in protecting riders from head injuries, which are often life-threatening. Governments across the world have implemented strict rules mandating helmet usage. Despite these regulations, many riders continue to ignore safety norms, leading to increased accident severity and fatalities. The main reason behind this issue is the lack of efficient and continuous monitoring systems.

Traditional traffic monitoring systems rely on manual observation by traffic police or basic CCTV cameras. These methods have several limitations, such as dependency on human effort, limited coverage, delayed response, and susceptibility to human error. In busy urban environments, it becomes extremely difficult for authorities to monitor every rider effectively. Moreover, reviewing CCTV footage manually is time-consuming and inefficient.

Recent advancements in Artificial Intelligence (AI) and Computer Vision have opened new possibilities for automating traffic monitoring systems. Deep learning techniques, particularly object detection algorithms, have shown significant success in identifying and classifying objects in images and videos. One such advanced algorithm is YOLOv8, which is capable of performing real-time object detection with high accuracy and speed.

This project focuses on developing an intelligent helmet detection system using YOLOv8. The system is designed to automatically detect two-wheeler riders and determine whether they are wearing helmets or not. By processing live video streams or recorded footage, the system can identify violations instantly and assist authorities in enforcing traffic rules more effectively.

The proposed solution aims to reduce dependency on manual monitoring, improve detection accuracy, and provide real-time alerts for violations. It can be integrated into existing traffic surveillance systems and scaled for use in smart city applications. By ensuring better enforcement of helmet laws, the system contributes to enhancing road safety and reducing accident-related fatalities.

1.2 Motivation

Road accidents are one of the leading causes of death worldwide, and a significant number of these fatalities involve two-wheeler riders. Studies have shown that wearing a helmet can reduce the risk of head injury by more than 60%. However, despite awareness campaigns and strict regulations, helmet compliance remains low in many regions.

The motivation behind this project is to develop an automated and intelligent system that can monitor helmet usage continuously without human intervention. By leveraging deep learning and computer vision technologies, the system aims to improve traffic law enforcement and promote safer riding practices.

1.3 Objectives

The main objectives of this project are:

To develop a real-time helmet detection system using YOLOv8

To detect two-wheeler riders from video streams or images

To classify riders as helmet or non-helmet users

To improve accuracy and efficiency compared to traditional systems

To assist traffic authorities in automated violation detection

1.4 Scope of the Project

The scope of this project includes:

Real-time monitoring using CCTV or video input

Detection of helmet usage using deep learning models

Storage of violation data for further processing

Integration with smart traffic systems in the future

CHAPTER 2

LITERATURE SURVEY

2.1 Introduction

Traffic rule violations, especially among two-wheeler riders, have become a major concern in modern urban environments. Helmet non-compliance and traffic signal violations significantly contribute to road accidents and fatalities. With the advancement of Artificial Intelligence (AI) and Computer Vision, several automated systems have been developed to detect and monitor such violations.

This chapter reviews various research works related to helmet detection and traffic violation monitoring systems. The focus is on deep learning-based approaches, particularly those using YOLO and Convolutional Neural Networks (CNN), which have shown promising results in real-time object detection tasks.

2.2 Automated Traffic Violation Detection Using YOLOv8 (Ayoub Charef et al., 2024)

Ayoub Charef et al. (2024) proposed an automated system for detecting helmet usage and traffic light violations using the YOLOv8 object detection algorithm. The study was conducted in Marrakech, Morocco, where video data was collected from traffic intersections using high-definition cameras.

The system utilizes a pretrained YOLOv8 model trained on the COCO dataset to detect objects such as motorcycles, persons, helmets, and traffic lights. The methodology involves processing video frames and identifying violations such as riders not wearing helmets and vehicles crossing signals during red lights.

The results of the study showed a strong correlation between automated detection and manual counting, indicating the reliability of the system. However, minor discrepancies were observed during high-speed traffic conditions, where fast-moving vehicles affected detection accuracy.

Advantages:

- Real-time detection capability
- High accuracy using YOLOv8
- Practical implementation using real-world data

Limitations:

- Reduced accuracy in high-speed scenarios
- Dependence on lighting and environmental conditions

2.3 Traffic Violation Detection Using Deep Learning (Ferdian et al., 2022)

Ferdian et al. developed a traffic violation detection system using deep learning techniques, focusing on identifying violations such as helmet absence and license plate recognition. The system uses object detection algorithms to analyze surveillance videos.

Their approach achieved high accuracy in detecting violations and demonstrated the effectiveness of deep learning models in traffic monitoring applications.

Advantages:

- High detection accuracy
- Multi-violation detection capability

Limitations:

- Computationally intensive
- Limited real-time performance

2.4 Helmet Violation Detection Using Computer Vision (Afzal et al., 2021)

Afzal et al. proposed a system for detecting helmet violations using deep learning and computer vision techniques. The system processes surveillance videos and identifies riders without helmets using trained models.

The results showed that the system performs efficiently in controlled environments and provides accurate detection results.

Advantages:

- Automated detection

- Reduces manual effort

Limitations:

- Performance affected by occlusion
- Limited scalability

2.5 CNN-Based Helmet Detection System (Dasgupta et al., 2019)

Dasgupta et al. developed a helmet detection system using Convolutional Neural Networks (CNN). The system focuses on detecting multiple riders and identifying whether they are wearing helmets.

The model achieved good accuracy but required significant computational resources and training time.

Advantages:

- Effective feature extraction using CNN
- Accurate classification

Limitations:

- Slow processing speed
- Not suitable for real-time applications

2.6 YOLOv3-Based Helmet Detection (Waris et al., 2022)

Waris et al. proposed a framework using YOLOv3 for detecting helmet violations. The system identifies motorcycles and riders and classifies helmet usage using deep learning techniques.

The system showed promising results in detecting multiple riders and violations in traffic videos.

Advantages:

- Faster detection compared to traditional methods
- Works well with multiple riders

Limitations:

- Lower accuracy compared to newer YOLO versions
- Limited adaptability to complex environments

2.7 Automated Motorcycle Violation Detection (Naik et al., 2021)

Naik et al. developed a real-time traffic rule violation detection system using

surveillance videos. The system detects helmet violations and other traffic offenses using deep learning models.

The results demonstrated improved efficiency in monitoring traffic violations compared to manual methods.

Advantages:

- Real-time monitoring
- Improved enforcement efficiency

Limitations:

- Requires high-quality video input
- Sensitive to environmental conditions

2.8 AI-Based Helmet Detection for Road Safety (Duong et al., 2023)

Duong et al. proposed an advanced system for detecting helmet violations using deep learning techniques. The system focuses on improving accuracy and robustness in real-world scenarios.

The model achieved good performance in detecting violations under varying conditions.

Advantages:

- Robust detection
- Handles complex scenarios

Limitations:

- Requires large datasets
- Computational complexity

2.9 Challenges Identified in Existing Systems

From the above studies, several challenges have been identified:

- **Lighting Conditions:** Poor lighting affects detection accuracy
- **High-Speed Vehicles:** Fast-moving vehicles reduce detection performance
- **Occlusion:** Objects blocking riders make detection difficult
- **Scalability Issues:** Some systems are not suitable for large-scale deployment
- **Real-Time Limitations:** Older models fail in real-time applications

2.10 Research Gap

Despite the advancements in traffic violation detection systems, several gaps still exist:

- Lack of highly accurate real-time detection systems
- Limited integration with smart city infrastructure
- Inconsistent performance under varying environmental conditions
- Dependence on outdated models like YOLOv3 and YOLOv5

2.11 Proposed Solution Based on Literature

Based on the analysis of existing systems, the proposed project uses YOLOv8, which offers:

- Higher accuracy
- Faster processing speed
- Better real-time performance
- Improved scalability

The system aims to overcome the limitations of previous models by providing a robust and efficient helmet detection solution.

2.12 Conclusion

The literature survey highlights the significant progress made in the field of automated traffic violation detection using AI and deep learning. Various approaches have been proposed using CNN and YOLO-based models, each contributing to improving road safety.

However, challenges such as accuracy, real-time processing, and scalability still exist. The proposed system using YOLOv8 addresses these challenges and provides an advanced solution for helmet detection and traffic monitoring.

Comparison Table of Existing Systems

S.No	Author & Year	Method Used	Dataset	Accuracy	Real-Time Capability	Advantages	Limitations
1	Ayoub Charef et al. (2024)	YOLOv8	Real-world traffic data	High	Yes	Real-time detection, high accuracy	Issues at high speed, lighting problems
2	Ferdian et al. (2022)	Deep Learning	Surveillance videos	High	Partial	Multi-violation detection	Computationally heavy
3	Afzal et al. (2021)	CNN + CV	Video data	Moderate	Limited	Automated detection	Occlusion issues
4	Dasgupta et al. (2019)	CNN	Image dataset	Good	No	Good classification accuracy	Slow processing
5	Waris et al. (2022)	YOLOv3	Traffic videos	Good	Yes	Faster than traditional methods	Lower accuracy than YOLOv8
6	Naik et al. (2021)	Deep Learning	Surveillance footage	Moderate	Yes	Real-time monitoring	Sensitive to video quality
7	Duong et al. (2023)	Deep Learning	Large dataset	High	Yes	Robust detection	High computational cost

CHAPTER 3

PROPOSED SYSTEM

3.1 Proposed System Overview

The proposed system focuses on developing an intelligent and automated helmet detection solution using advanced deep learning techniques. The primary objective of this system is to identify whether two-wheeler riders are wearing helmets or not in real-time using images, video streams, or CCTV footage. The system is designed to overcome the limitations of traditional monitoring methods by providing a fully automated, accurate, and scalable solution for traffic rule enforcement.

The system leverages the power of the YOLOv8 (You Only Look Once version 8) object detection framework, which is known for its high speed and accuracy in detecting objects in real-time. By utilizing deep learning and computer vision technologies, the system ensures efficient monitoring of traffic violations without requiring continuous human intervention.

3.2 Dataset Collection and Preparation

A large-scale dataset is used to train the model, consisting of more than 9,000 images collected from multiple publicly available helmet detection datasets. The dataset includes diverse scenarios such as:

- Riders wearing helmets
- Riders without helmets
- Different lighting conditions
- Various traffic environments
- Multiple rider positions and angles

To ensure high model performance, the dataset undergoes several preprocessing steps. These include image resizing, normalization, annotation, and augmentation. The images are structured in YOLO format, where each image is associated with bounding box coordinates and class labels.

Data augmentation techniques such as rotation, flipping, zooming, and brightness adjustment are applied to improve the model's robustness and generalization capability. This helps the system perform well even in challenging real-world conditions.

3.3 Model Selection and Training

The system uses the YOLOv8 medium architecture (YOLOv8m), which provides an optimal balance between detection accuracy and processing speed. This makes it suitable for real-time applications such as traffic monitoring systems.

The model is trained using GPU acceleration to handle large datasets efficiently and reduce training time. During training, the system optimizes key performance metrics such as:

- Mean Average Precision (mAP)
- Precision
- Recall
- F1 Score

The training process involves multiple epochs, where the model learns to identify patterns and features associated with helmets and riders. Loss functions are minimized during training to improve prediction accuracy.

3.4 System Functionality

Once the model is trained, the system performs the following key functions:

1. **Real-Time Detection** The system detects two-wheeler riders and identifies whether they are wearing helmets in real time.
2. **Video Processing** It processes live camera feeds or recorded CCTV footage frame by frame to detect violations.
3. **Automated Violation Detection** Riders without helmets are automatically identified and flagged as violators.
4. **Smart Monitoring Support** The system supports integration with smart traffic management systems for efficient monitoring. YOLOv8-Based Object Detection

The core of the proposed system is the YOLOv8 object detection algorithm developed by Ultralytics. YOLOv8 is a single-stage detector that processes the entire image in one pass, making it extremely fast and efficient.

The detection pipeline works as follows:

1. Capture video input from CCTV or camera
2. Convert video into frames
3. Pass frames into YOLOv8 model
4. Detect objects such as:
    a. Motorcycles
    b. Riders
    c. Helmets
5. Classify riders as:
    a. Helmet
    b. No Helmet

The model draws bounding boxes around detected objects and labels them accordingly. This enables easy visualization of results.

3.5 Real-Time Monitoring System

The proposed system includes a real-time monitoring dashboard that displays detection results instantly. This dashboard provides:

- Live video feed with detection overlays
- Highlighting of helmet and non-helmet riders
- Automatic alerts for violations
- Capture of offender images
- Timestamp logging

Whenever a rider without a helmet is detected, the system immediately captures the frame and stores it as evidence. This helps authorities take quick action against violators.

3.6 Machine Learning Integration

Machine learning plays a crucial role in enhancing the system's performance and intelligence. The system is capable of:

- Learning from large datasets
- Improving accuracy over time
- Analyzing traffic patterns
- Tracking frequent violations
- Supporting predictive analysis

By analyzing historical data, the system can identify high-risk areas and time periods where violations are more frequent. This information can be used for better traffic planning and enforcement strategies.

3.7 Security and Data Management

The system ensures secure handling and storage of data. Since the system deals with sensitive information such as violation records, proper security mechanisms are implemented.

Key features include:

- Secure database storage
- Role-based access control
- Data encryption
- Backup and recovery mechanisms
- Automated report generation

Only authorized personnel can access violation data, ensuring privacy and data protection.

3.8 Scalability and Deployment

The proposed system is highly scalable and can be deployed in various environments such as:

- Highways
- Traffic intersections
- Urban roads
- College campuses
- Smart cities

The system can be integrated with existing surveillance infrastructure and expanded to cover large areas. It can also be connected to automated fine (e-challan) systems in the future.

3.9 Modules of the Proposed System

The system is divided into multiple modules for efficient functioning:

1. User Interface Module

This module provides a graphical interface for users to interact with the system. It displays detection results, alerts, and reports in an easy-to-understand format.

2. Data Collection Module

This module captures video input from CCTV cameras or live streams. It ensures continuous data flow for processing.

3. Preprocessing Module

This module processes raw data by:

- Extracting frames
- Cleaning images
- Resizing and normalizing data

YOLOv8 Detection Module

This is the core module where object detection takes place. It identifies riders and helmets using the trained YOLOv8 model.

5. Violation Detection Module

This module checks whether the detected rider is wearing a helmet or not. If not, it flags the rider as a violator. 6. Database and Security Module

This module stores violation data securely and manages access control. It also generates reports for authorities.

Advantages of the Proposed System

- Fully automated system
- High accuracy and speed
- Real-time detection capability
- Reduced human effort
- Scalable and flexible
- Easy integration with smart systems

3.10 Conclusion of Proposed System

The proposed system provides an efficient and intelligent solution for helmet detection using advanced deep learning techniques. By automating the process of traffic monitoring, it significantly reduces manual effort and improves enforcement efficiency. The use of YOLOv8 ensures high accuracy and real-time performance, making the system suitable for practical deployment in modern smart city environments.

CHAPTER 4

DESIGN AND MODELLING OF SYSTEM.

Unified Modelling Language (UML) is a standardized modeling language used to visualize, specify, construct, and document the components of a system. It acts as a blueprint for designing software systems by representing their structure and behavior.

In this project, UML diagrams are used to describe the architecture and workflow of the **Helmet Detection System using YOLOv8**. These diagrams help in understanding how different components of the system interact with each other and how data flows through the system.

The following UML diagrams have been designed for the proposed system:

1. Use Case Diagram
2. Class Diagram
3. Sequence Diagram
4. Activity Diagram
5. State Diagram

4.1 Use Case Diagram

A Use Case Diagram is a behavioral diagram that represents the interactions between users (actors) and the system. It shows the functionalities provided by the system and how different users utilize those functionalities.

In the proposed helmet detection system, the main actors involved are:

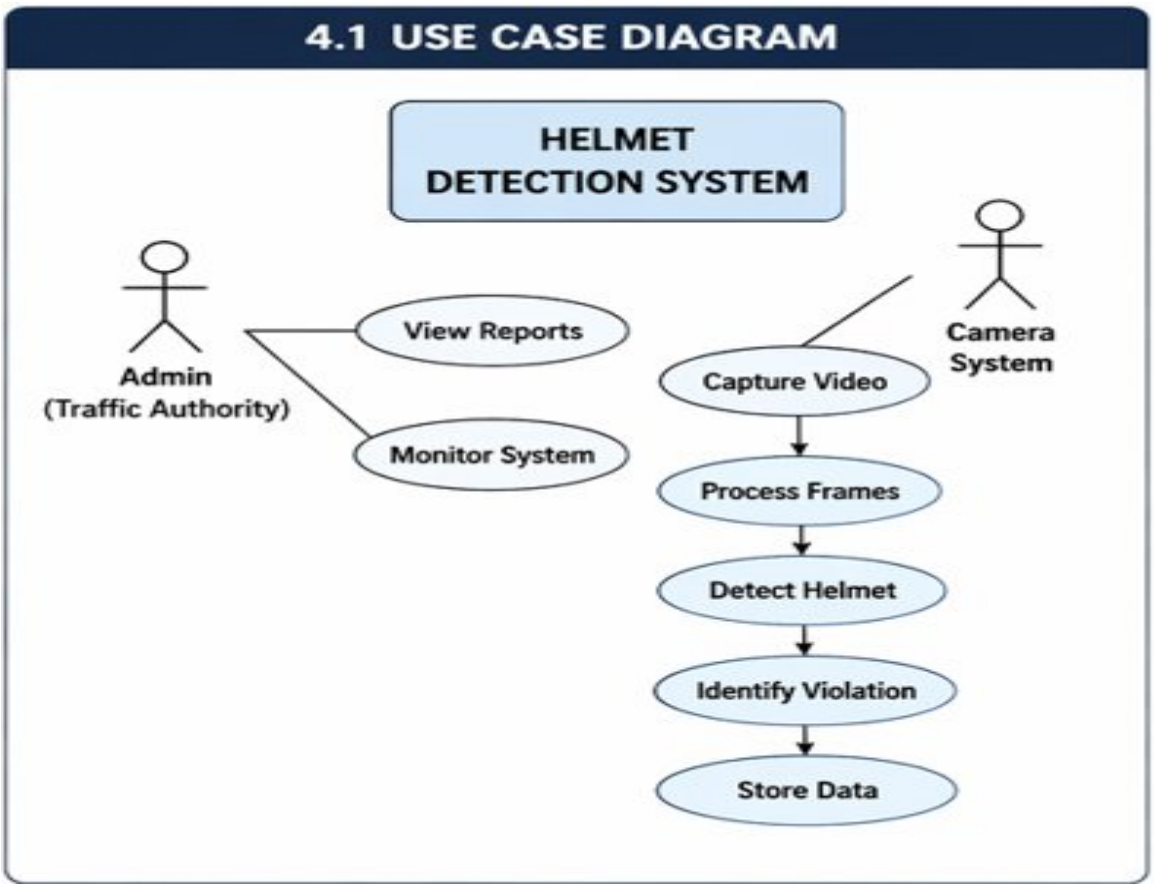
- **Admin (Traffic Authority)**
- **Camera System (CCTV/Input Device)**
- **System (Helmet Detection Model)**

Use Cases:

- Capture Video Stream

- Process Video Frames
- Detect Motorcycles
- Detect Helmet / No Helmet
- Identify Violations
- Store Violation Data
- Generate Reports

The system captures live video from CCTV cameras, processes the frames using the YOLOv8 model, detects riders and helmets, and identifies violations. The results are stored in a database and can be accessed by the admin.



4.2 Class Diagram

A Class Diagram represents the structure of the system by showing different classes, their attributes, methods, and relationships.

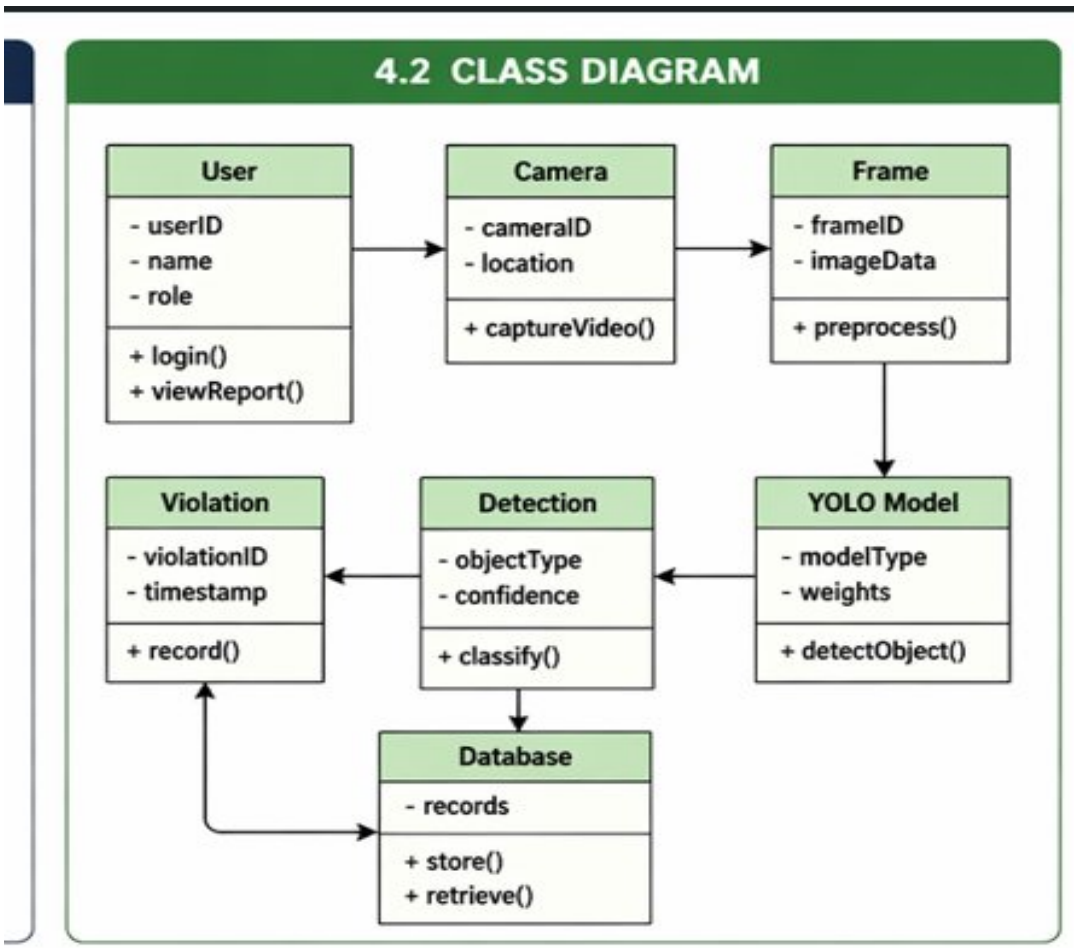
Main Classes in the System:

- **User (Admin)**
  - Attributes: userID, name, role
  - Methods: login(), viewReports()
- **Camera**

    - Attributes: cameraID, location
    - Methods: captureVideo()
- **Frame**
    - Attributes: frameID, imageData
    - Methods: preprocess()
- **YOLOModel**
    - Attributes: modelType, weights
    - Methods: detectObjects()
- **Detection**
    - Attributes: objectType, confidenceScore
    - Methods: classifyHelmet()
- **Violation**
    - Attributes: violationID, timestamp
    - Methods: recordViolation()
- **Database**
    - Attributes: records
    - Methods: storeData(), retrieveData()

Relationships:

- Camera → provides frames
- YOLOModel → processes frames
- Detection → identifies helmet status
- Violation → stores detected violations



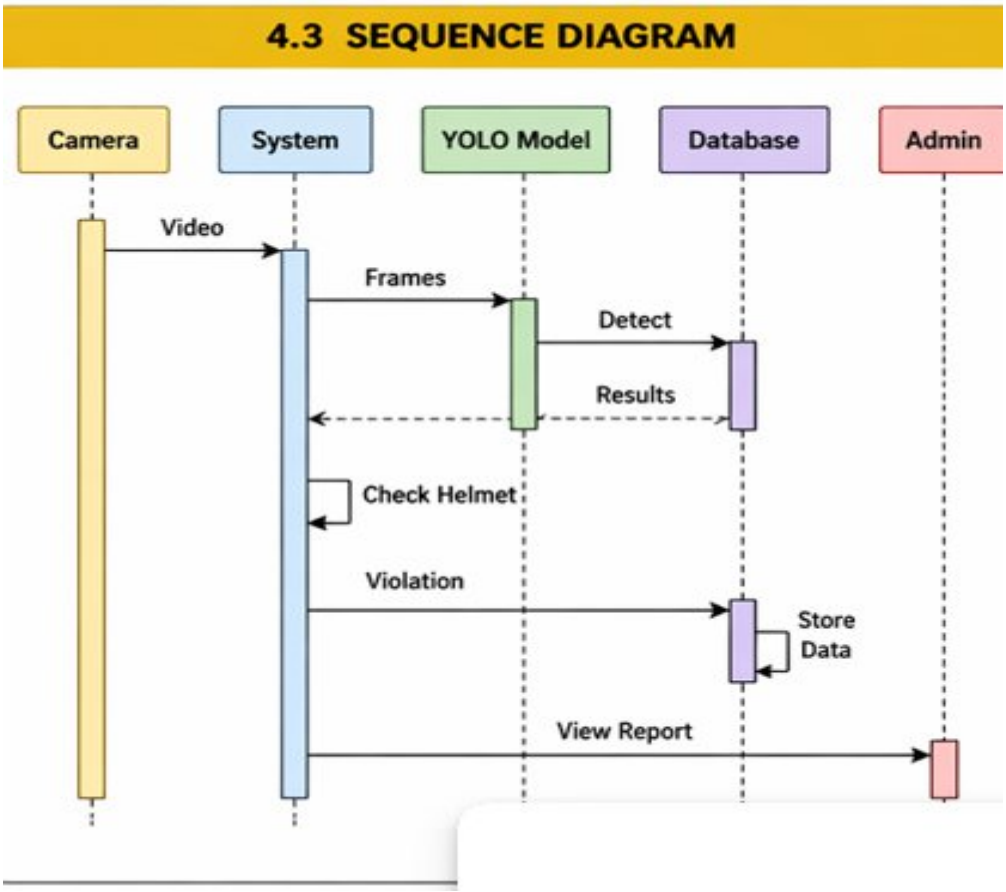
4.3 Sequence Diagram

A Sequence Diagram shows the step-by-step interaction between system components in a time sequence.

Flow of Events:

1. Camera captures video
2. Frames are extracted
3. Frames sent to YOLOv8 model
4. Model detects objects (rider, helmet)
5. System checks helmet presence
6. If no helmet → violation generated
7. Data stored in database
8. Admin views results

This diagram clearly shows how data flows from input to output in a sequential manner.



4.4 Activity Diagram

An Activity Diagram represents the workflow of the system from start to end.

Flow:

Start

↓

Capture video

↓

Extract frames

↓

Preprocess frames

↓

Run YOLOv8 detection

↓

Is helmet detected? → Yes → Continue monitoring

→ No → Record violation

↓

Store data

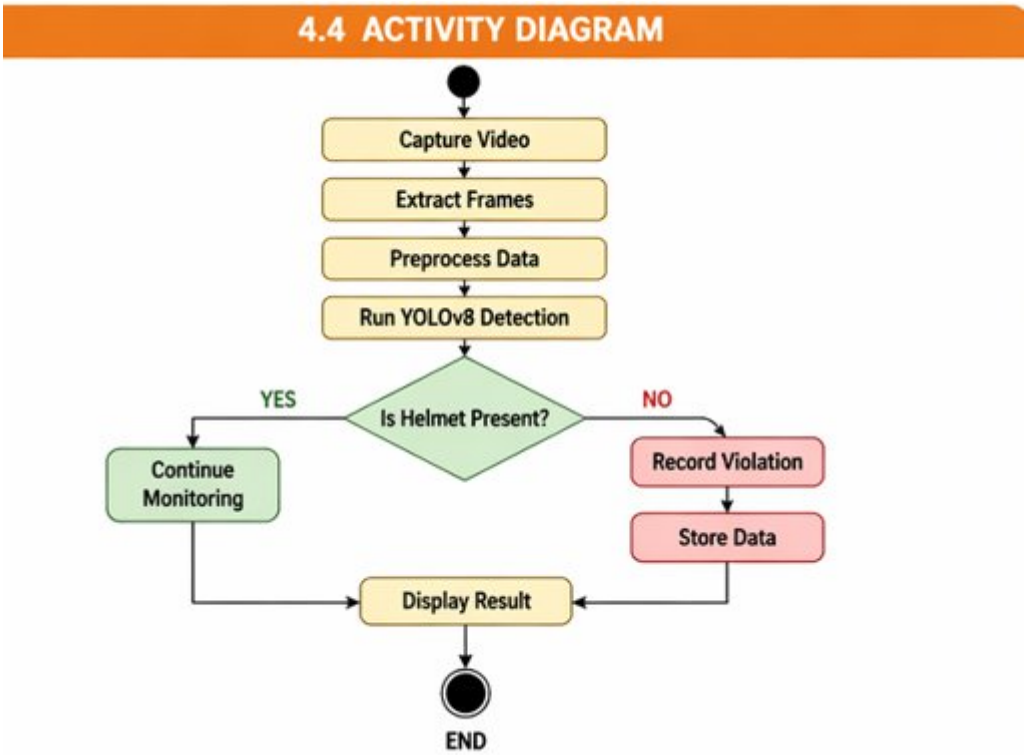
↓

Display results

↓

End

This diagram helps visualize the decision-making process in the system.



4.5 State Diagram

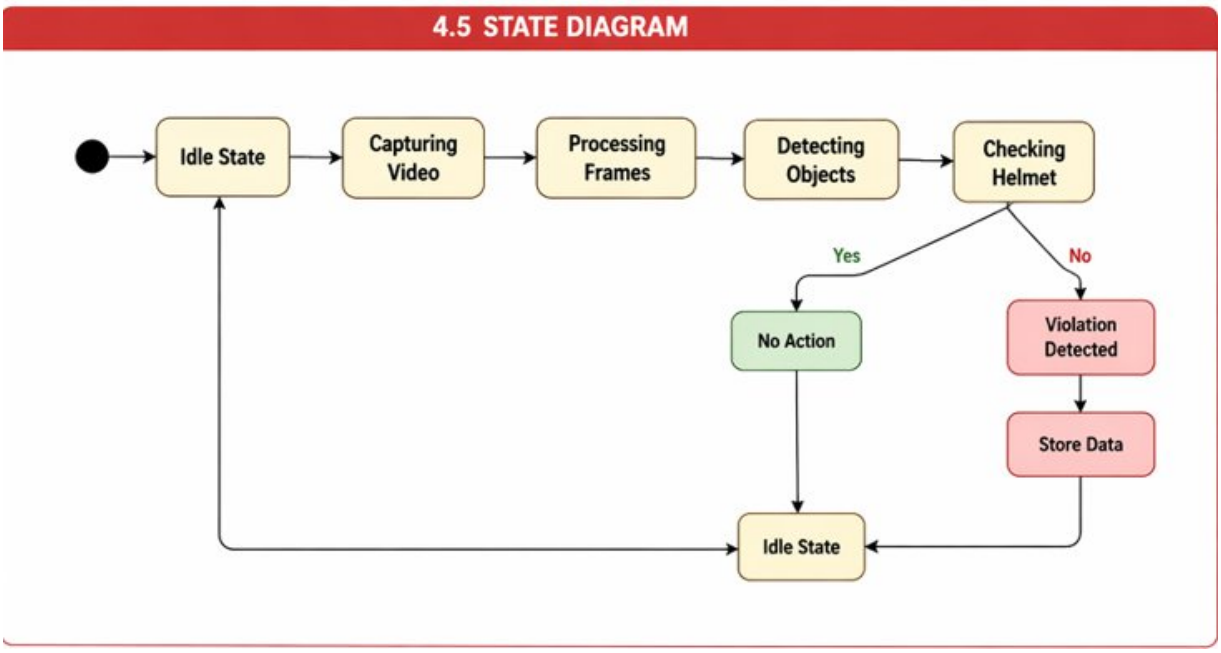
A State Diagram represents different states of the system and transitions between them.

States of the System:

- Idle State
- Capturing State
- Processing State
- Detection State
- Violation State
- Storage State

Transitions:

- Idle → Capturing (when camera starts)
- Capturing → Processing (frame extraction)
- Processing → Detection (model runs)
- Detection → Violation (if no helmet)
- Detection → Idle (if helmet present)



CHAPTER 5

ALGORITHM AND METHODS

5.1 Data Acquisition

Data acquisition is a crucial step in developing an accurate and reliable helmet detection system. In this project, a large-scale dataset consisting of more than **9,000 images** was used to train and evaluate the performance of the YOLOv8 model.

The dataset was collected from multiple publicly available sources such as open datasets and online repositories. It includes images of:

- Two-wheeler riders wearing helmets
- Riders without helmets
- Different traffic environments
- Various lighting conditions (day/night)
- Multiple viewing angles and camera positions

To ensure better generalization, the dataset contains images captured from real-world scenarios such as highways, city roads, and intersections.

The dataset is organized into three main folders:

- **Training Set** – used to train the model
- **Validation Set** – used to tune model parameters
- **Testing Set** – used to evaluate model performance

Each image is annotated in **YOLO format**, which includes:

- Class label (helmet / no helmet)
- Bounding box coordinates

These annotations help the model learn object localization and classification effectively.

5.2 Data Pre-processing

Data preprocessing is an essential step to improve the quality of input data and enhance model performance. The following preprocessing techniques were applied:

1. Image Resizing

All images were resized to a fixed resolution (e.g., 640×640 pixels) to ensure consistency during training.

2. Normalization

Pixel values were scaled between 0 and 1 using:

$$ \text{Normalized Value} = \frac{\text{Pixel Value}}{255} $$

This helps in faster convergence during training.

3. Data Augmentation

To improve robustness and avoid overfitting, various augmentation techniques were applied:

- Rotation
- Horizontal flipping
- Zooming
- Brightness adjustment
- Shearing

Preprocessing Parameters

Technique	Value
Rescale	1/255
Zoom Range	0.2
Shear Range	0.2
Horizontal Flip	True

These techniques allow the model to perform well under different environmental conditions.

5.3 Proposed Model: YOLOv8 Architecture

In this project, the **YOLOv8 (You Only Look Once version 8)** model is used for object detection. YOLOv8 is a state-of-the-art deep learning model designed for real-time detection tasks.

Why YOLOv8?

- High accuracy
- Fast processing speed
- Real-time detection capability
- Improved feature extraction

The **YOLOv8 medium model (YOLOv8m)** is selected to balance performance and speed.

5.4 Working of YOLOv8 Algorithm

YOLOv8 is a single-stage object detection algorithm that processes the entire image in one pass. Unlike traditional methods, it does not require separate region proposal steps.

Steps Involved:

1. Input image is passed to the model
2. Image is divided into grids
3. Model predicts:
    a. Bounding boxes
    b. Class probabilities
4. Non-Max Suppression (NMS) is applied to remove duplicate detections
5. Final output displays detected objects

5.5 Mathematical Representation

Bounding Box Prediction

Each bounding box is represented as:

[(x, y, w, h)]

Where:

- (x, y) = center coordinates
- (w, h) = width and height

Activation Function (ReLU)

$$f(x) = \max(0, x)$$

Used to introduce non-linearity in the network.

Sigmoid Function

$$S(x) = \frac{1}{1 + e^{-x}}$$

Used for probability prediction.

Softmax Function

Used for multi-class classification:

$$P(y_i) = \frac{e^{z_i}}{\sum e^{z_j}}$$

5.6 Training Process

The model is trained using:

- **GPU acceleration** for faster computation
- Multiple epochs to improve learning
- Optimizer (Adam/SGD)
- Loss function to minimize prediction error

Evaluation Metrics:

- Accuracy
- Precision
- Recall
- Mean Average Precision (mAP)

5.7 Helmet Detection Algorithm

Algorithm Steps:

1. Capture video from camera
2. Extract frames from video
3. Preprocess frames
4. Pass frames to YOLOv8 model
5. Detect objects (motorcycle, rider, helmet)
6. Check helmet presence
7. If no helmet:
    a. Mark as violation
    b. Store image
8. Display results

5.8 Flow of Algorithm

```
Start
 ↓
Capture Video
 ↓
Extract Frames
 ↓
Preprocess Data
 ↓
Run YOLOv8 Model
 ↓
Detect Helmet?
   /     \
 Yes     No
  |       |
 Continue  Record Violation
  |       |
  ↓       ↓
 Display  Store Data
  ↓       ↓
       End
```

5.9 Advantages of Proposed Method

- Real-time detection
- High accuracy
- Automated system
- Scalable for smart cities
- Reduced human intervention

5.10 Conclusion

The proposed algorithm using YOLOv8 provides an efficient and reliable solution for helmet detection. By combining deep learning with real-time processing, the system ensures accurate detection of traffic violations and contributes to improving road safety.

CHAPTER 6

IMPLEMENTATION AND TESTING

6.1 Data Acquisition and Environment Setup

The implementation of the proposed helmet detection system was carried out using **Google Colab**, which provides a cloud-based environment with GPU support for training deep learning models.

The dataset used for this project was obtained from publicly available sources and consists of more than **9,000 images** of two-wheeler riders with and without helmets. The dataset was downloaded from online repositories and imported into the Colab environment using Kaggle API.

Steps Involved:

1. **Setting up Kaggle API in Google Colab**
    a. Upload Kaggle API key
    b. Install Kaggle library
2. **Downloading Dataset**
    a. Dataset downloaded using Kaggle commands

3. **Extracting Dataset**
    a. The dataset is extracted into training, validation, and testing folders
4. **Directory Structure**
    a. Train folder
    b. Validation folder
    c. Test folder

6.2 Data Preprocessing and Augmentation

To improve model performance and ensure better generalization, data preprocessing and augmentation techniques were applied.

Preprocessing Steps:

- Image resizing (e.g., 640×640)
- Normalization (scaling pixel values)
- Conversion into YOLO format

Data Augmentation Techniques:

- Horizontal flipping
- Rotation
- Zooming
- Brightness adjustment
- Shearing

These techniques help the model learn from varied conditions and reduce overfitting.

Implementation

Instead of using TensorFlow-based generators, YOLOv8 internally handles augmentation during training using its built-in pipeline.

6.3 Model Implementation (YOLOv8)

The helmet detection system is implemented using the **YOLOv8 object detection model** provided by Ultralytics.

Steps for Implementation:

1. Install YOLOv8 library
2. Load pretrained YOLOv8 model
3. Configure dataset YAML file

4. Train the model using custom dataset
5. Validate model performance
6. Run inference on test data

Training Command (Example):

```
yolo detect train data=data.yaml model=yolov8m.pt epochs=50 imgsz=640
```

Model Used:

- YOLOv8 Medium (YOLOv8m)

Training Parameters:

- Epochs: 50
- Batch size: 16 or 32
- Image size: 640

6.4 Testing and Detection

After training, the model is tested using:

- Test images
- Real-time video input
- CCTV footage

Testing Process:

1. Input image/video
2. Model processes input
3. Detect objects (rider, helmet)
4. Display bounding boxes
5. Classify as:
    a. Helmet
    b. No Helmet

Output:

- Bounding boxes around detected objects
- Labels with confidence scores

6.5 Performance Evaluation

The performance of the YOLOv8 model is evaluated using the following metrics:

- **Precision** – correctness of predictions
- **Recall** – ability to detect all objects
- **Accuracy** – overall performance
- **mAP (Mean Average Precision)** – standard object detection metric

The model achieved high accuracy in detecting helmet and non-helmet riders in real-time scenarios.

6.6 Comparison with Other Models

Unlike traditional CNN models such as AlexNet, VGG16, and ResNet, YOLOv8 provides:

- Faster detection speed
- Real-time performance
- Better object localization
- Improved accuracy

YOLOv8 outperforms older models in real-time applications, making it more suitable for traffic monitoring systems.

6.7 Results Visualization

The results are visualized using:

- Detection images
- Bounding boxes
- Confidence scores

```python
from IPython.display import Image
Image("runs/detect/predict-6/test(2).jpg")
```

```python
from IPython.display import Image
Image("runs/detect/predict-2/test(1).jpg")
```

CHAPTER 7

EVALUATION

7.1 Data Preprocessing and Augmentation Analysis

To improve the performance and robustness of the helmet detection model, data preprocessing and augmentation techniques were applied. These techniques help the model generalize better and perform accurately on unseen data.

Data augmentation artificially increases the dataset size by generating modified versions of existing images. This improves the model's ability to handle real-world variations such as lighting conditions, camera angles, and motion blur.

Augmentation Techniques Used:

- **Rescaling:** Pixel values are scaled from the range [0–255] to [0–1] to normalize the data and improve training stability.
- **Rotation:** Images are randomly rotated to simulate different camera angles.
- **Horizontal Flip:** Images are flipped horizontally to increase variation.
- **Zooming:** Images are zoomed in or out to simulate different distances.
- **Brightness Adjustment:** Adjusts lighting conditions for better robustness in day/night scenarios.
- **Shearing:** Applies geometric transformation to simulate different perspectives.

Impact of Augmentation:

- Improves model accuracy
- Reduces overfitting
- Enhances detection under different conditions

7.2 Performance Metrics

The performance of the YOLOv8 model is evaluated using standard object detection metrics:

1. Precision

Precision measures how many of the detected helmets are actually correct.

[Precision = \frac{TP}{TP + FP}]

2. Recall

Recall measures how many actual helmets were correctly detected.

[Recall = \frac{TP}{TP + FN}]

3. Accuracy

Accuracy represents the overall correctness of the model.

4. Mean Average Precision (mAP)

mAP is the most important metric for object detection models and measures detection accuracy across all classes.

7.3 Confusion Matrix Analysis

A confusion matrix is used to evaluate the classification performance of the model.

For helmet detection, the confusion matrix consists of:

- **True Positive (TP):** Helmet correctly detected
- **True Negative (TN):** No-helmet correctly detected
- **False Positive (FP):** Helmet incorrectly detected
- **False Negative (FN):** Helmet missed

Interpretation:

- High TP and TN values indicate good performance
- Low FP and FN values indicate fewer errors
- The model shows strong classification capability for helmet detection

7.4 Detection Results and Visualization

The trained YOLOv8 model is tested on various images and video inputs. The model successfully detects:

- Two-wheeler riders
- Helmets
- Non-helmet violations

Each detection is displayed with:

- Bounding boxes
- Class labels (Helmet / No Helmet)
- Confidence scores (%)

Observations:

- High accuracy in detecting helmets
- Works well in real-time scenarios
- Minor errors in low-light or occluded conditions

7.5 Real-Time Testing

The system was tested using:

- Live camera feed
- CCTV footage
- Recorded videos

Results:

- Real-time detection achieved
- Fast processing speed
- Accurate violation identification

The system successfully identifies riders without helmets and marks them as violators.

7.6 User Interface (Streamlit Application)

A simple and interactive user interface was developed using the Streamlit framework.

Streamlit allows easy deployment of machine learning models into web applications. The interface enables users to:

- Upload images or videos
- View detection results
- Display bounding boxes and labels
- Monitor violations

Features of UI:

- Simple and user-friendly
- Real-time output display

- Easy integration with model

7.7 Discussion of Results

The YOLOv8 model demonstrates strong performance in detecting helmet usage. The results indicate that:

- The model achieves high accuracy and precision
- Real-time detection is possible
- System performs well in most traffic conditions

Limitations:

- Reduced accuracy in low lighting
- Difficulty in detecting helmets in crowded scenes
- High-speed motion may affect detection

7.8 Conclusion

The evaluation results show that the proposed helmet detection system is efficient, accurate, and suitable for real-time applications. The use of YOLOv8 significantly improves detection performance compared to traditional methods.

The system can be effectively used in traffic monitoring and smart city applications to improve road safety.

CHAPTER 8

PUBLICATION

OBJECT DETECTION USING IMAGE PROCESSING

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Abstract— The use of two-wheelers has been increasing rapidly. But many two wheeler riders are not responsible for their safety as they are not wearing helmets. Manually Checking Whether Everyone Is Wearing A Helmet Is A Not Possible Especially In A Dense Urban Environment. By using the YOLOv8 framework, the helmet detection system is proposed to avoid this condition in real-time.

The proposed system is built for a large merged dataset consisting of over 9000 total images. Images utilized in the dataset are taken from different helmet detection datasets. The key purpose of using such a gigantic and merged dataset is that it can make the model more robust and generalize it for never witnessed data. In other words, the lack of a merged dataset can make us vulnerable to the overfitting problem. In addition.

Yolov5 and OpenCV are used for helmet detection. The complete method of this paper has been shown in figure 1 of the paper. Resizing of image is performed for better feature extraction. The algorithm relies on a weighted version of k-nearest neighbors. The mask's input sample is derived from skin colour samples taken from the skin.

Authorities have a serious problem of enforcing traffic rules which lead to accidents and causalities. To help combat this problem, the computer vision research group has designed an automated traffic monitoring and violation detection system. A vast number of smart cities are installing surveillance cameras to control the traffic on the roads. The responses pertain to the surveillance system.The solution offered may be.

I. INTRODUCTION

Motorcycles are popular throughout the world at this point. They provide twin advantages of price and convenience. Besides that, they also take part in a huge number of road accidents. Most deaths due to motorcycle accidents happen because of head injuries. A helmet must be used by every motorbike rider. This significantly reduces the likelihood of serious or grave injuries.

Wearing a good quality helmet may save your life in the event of a crash.

Processing and learning methods like objects' detection are used in automated monitoring for safety. As CCTV cameras and surveillance systems get deployed day by day, the chance for this implementation is getting increased day by day. Currently, image and video object detection models can identify real-time violations and detect any sort of infringement. A model from the YOLO family is one such model.

This paper proposes a real-time helmet detection system to automatically detect helmets on two-wheeler riders using YOLOv8 framework. Since benchmark helmet detection data is not available, the model is trained on a merged dataset of >9000 annotated images to make the model robust under varying conditions. The technique obtains a balance between detection accuracy and inference speed using medium architecture YOLOv8. The proposed method.

It also generates images and records videos of the activities. In addition, it also processes a live CCTV stream, ideal for intelligent traffic monitoring and smart city surveillance systems.

A. METHODOLOGY

1) Data Preparation.

Combining publicly available datasets to form a large dataset.A more varied dataset has also been created. The combined data is more than 9000 images The combined dataset contains pictures of. •. Cyclists wearing a helmet •.Model efficiency is enhanced by preprocessing the dataset with image resizing, normalization, annotation correction, and class balancing.

The YOLO annotation formatconversion took place for all the images. Following conversion, the images were divided into three sets namely training.

2) ModelSelection.

The team has selected the YOLOv8 Medium (YOLOv8m) architecture as it is 8 times smaller than the 8 billion parameter large language model. Consequently, it speeds up and fortifies detection.

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We have applied for publication of paper and its still pro**cessing**

CHAPTER 9

CONCLUSION

8.1 Conclusion

In this project, an intelligent and automated helmet detection system has been successfully developed using the YOLOv8 object detection algorithm. The primary objective of the system was to detect whether two-wheeler riders are wearing helmets or not in real-time using images and video streams. This system addresses the growing need for efficient traffic monitoring and enforcement of safety regulations.

The proposed system utilizes a large dataset of helmet and non-helmet images, which was preprocessed and used to train the YOLOv8 model. The model demonstrated high accuracy in detecting helmets and identifying violations. The use of deep learning and computer vision techniques enabled the system to perform real-time detection with minimal delay.

The implementation of the system in a cloud-based environment with GPU support ensured efficient training and faster processing. The integration of a user-friendly interface using Streamlit further enhances usability by allowing users to visualize detection results easily.

The evaluation results show that the system performs well under various conditions and is capable of detecting helmet violations accurately. Although minor limitations exist, such as reduced performance in low-light conditions and high-speed scenarios, the overall system proves to be effective and reliable.

This project demonstrates how modern AI-based technologies can be applied to real-world problems such as road safety and traffic management. The proposed solution reduces dependency on manual monitoring and provides a scalable approach for enforcing traffic rules.

8.2 Future Scope

Although the proposed system achieves good performance, there are several areas where further improvements can be made to enhance its functionality and applicability.

1. Number Plate Detection Integration

The system can be extended to include automatic number plate recognition (ANPR) to identify violators and generate e-challans automatically.

2. Real-Time Deployment in Smart Cities

The system can be deployed on a large scale in smart city infrastructures, including highways, intersections, and urban roads.

3. Improved Night-Time Detection

Advanced techniques such as infrared imaging and low-light enhancement can be used to improve performance in night conditions.

4. Multi-Violation Detection

The system can be enhanced to detect other traffic violations such as:

- Triple riding
- Over-speeding
- Signal jumping

5. Cloud Integration

Storing violation data on cloud platforms can enable centralized monitoring and better data management.

6. Mobile Application Development

A mobile application can be developed for traffic authorities to monitor violations in real time and receive alerts.

7. Model Optimization

Further optimization of the YOLOv8 model can reduce computational requirements and improve efficiency for edge devices.

8.3 Final Remarks

The proposed helmet detection system provides a reliable, efficient, and scalable solution for improving road safety. With further enhancements and real-world deployment, it has the potential to significantly reduce traffic violations and contribute to safer transportation systems.

CHAPTER 10

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